

# Numbers that regress

Overflow is demand above capacity

# The idea

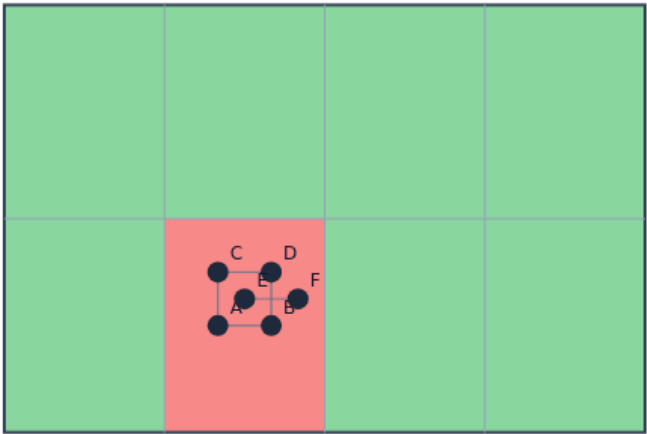
- Ov equals max of zero and demand minus Cap, per tile
- Total is the sum
- Max is the worst tile
- Count is how many tiles have positive overflow
- Report all three every time you change placement

# Define overflow

OVERFLOW METRICS

## Define overflow

Step 1 / 5 · def · module02-07-overflow-metrics



### Explanation

$ov = \max(0, \text{demand} - \text{Cap})$  per tile.

- Never negative

Cap=2

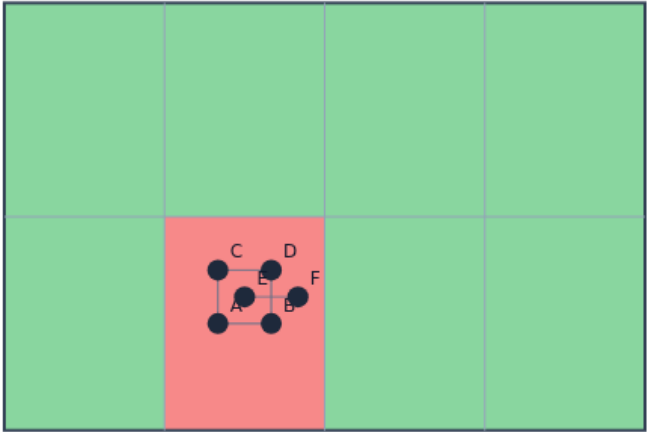
Define overflow

# Total overflow

OVERFLOW METRICS

Total overflow

Step 2 / 5 · total · module02-07-overflow-metrics



Explanation

Sum of per-tile overflow—primary regression number.

- Seed total = 5

Spread total higher but flatter

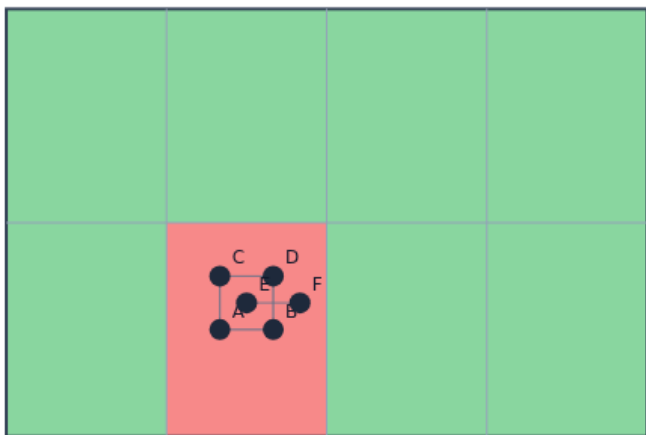
Total overflow

# Max overflow

OVERFLOW METRICS

## Max overflow

Step 3 / 5 · max · module02-07-overflow-metrics



### Explanation

Worst tile—catches hotspots even when total is moderate.

- Seed max = 5

Hotspot detector

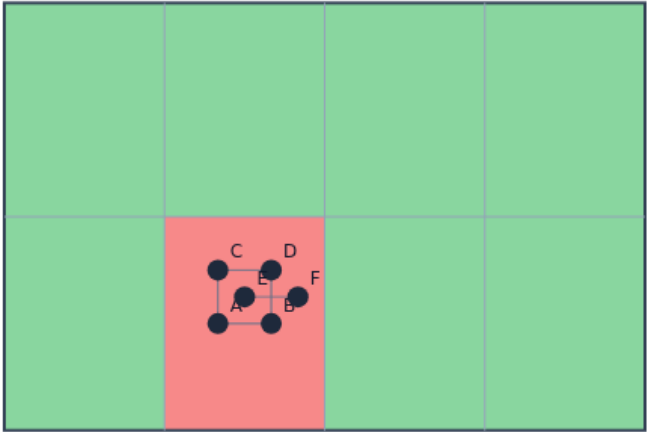
Max overflow

# Congested count

OVERFLOW METRICS

Congested count

Step 4 / 5 · count · module02-07-overflow-metrics



Explanation

How many tiles overflow—useful for “how widespread”.

- count on seed = 1

Triple report

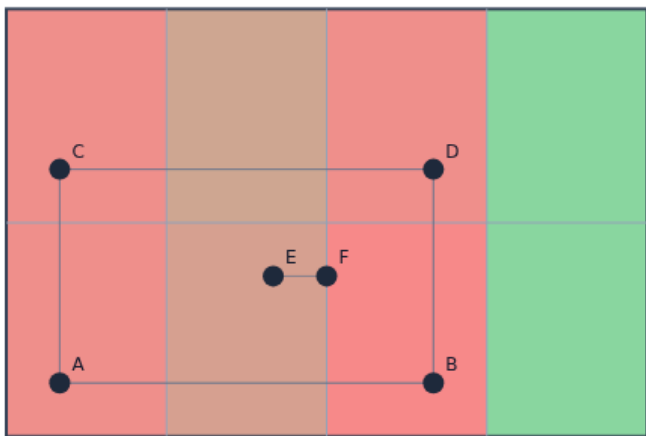
Congested count

# Hit a target

OVERFLOW METRICS

## Hit a target

Step 5 / 5 · target · module02-07-overflow-metrics



### Explanation

Move cells until total/max/count clear challenge thresholds.

- Check scores positions

No reveal required

Hit a target

# Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Overflow metrics

INTERACTIVE TOOL

## Overflow metrics

Report total, max, and congested GCell count from a demand map.

**Your job:** hit overflow targets by moving cells. Metrics: total · max · count.

Reset workspace

Challenges0 / 8 cleared

Pick one [Intro] Seed total > 0

Pick a challenge and click **Start**. Move cells, then **Check**. Reveal golden is study-only.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A B C D E F ← → ↑ ↓

Reset to starter

Reveal golden (study)

Run feedback

Graph

Reference / metrics

selected: A @ GCell (1,0)  
HPWL: 7.0  
overflow total: 5.00 · max: 5.00 · count: 1  
capacity: 2  
view: your placement

Browser lab starter

learn\_congestion — 07 overflow metrics



# Implement track

- Implement `overflow_metrics``
- Assert `congested_seed` has higher total overflow than spread placement at Cap equals two
- Print the triple (total, max, count)

# Pitfalls

- Reporting negative “overflow.” Counting tiles with congestion greater than one while
- Comparing totals across estimators with incompatible demand units

# Your turn

- Hit the overflow targets
- Next: cell inflation, the first feedback knob